

FINAL PROJECT
ME 5310
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OBJECTIVE:

The smooth, round finish of most shafts or bores makes them hard to attach to parts but retaining rings clamp around them and create a shoulder to hold pieces. In the current problem an initial geometry is provided for a retaining ring for which opening displacement can be calculated from the given formula $0.125 + ID * 0.125$ inch. The ID was given for individuals to get their own unique analysis. Since the ring is rarely inserted and removed, the main concern is material yielding, which loosens the fit. To choose suitable material for a ring, the von Mises effective stress is considered as a metric of material yielding.

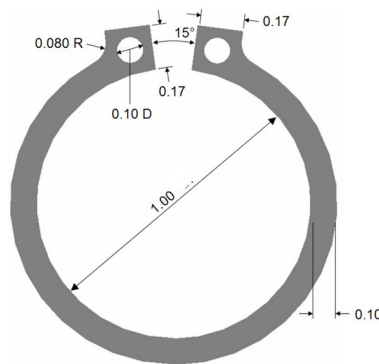


Figure 01: Given retaining ring geometry with dimensions in inches

ASSUMPTIONS:

- Each of the given dimensions is expressed in inches. To facilitate our analysis, the units have been transformed into SI units.
- The analyses assumed a symmetric ring opening, with each hole displaced by the same distance in opposite directions, equal to half of the total opening distance.

MODELLING THE RETAINING RING:

The ABAQUS model of the ring conforms to the specifications outlined in the design statement. The sketch below illustrates the completed model constructed with ABAQUS.



Figure 02: Model of the ring in ABAQUS created with dimensions in millimeters

DISPLACEMENT BOUNDARY CONDITION:

Given, ID = 0.336681781
Displacement Formula, $d = 0.125 + ID \cdot 0.125$
 $= 0.167085 \text{ Inch}$
 $= 4.243965 \text{ mm}$

This is the distance at which the ring must open which is determined by the center-to-center distance of the holes. Following the coupling of the center points of the holes with the interior surfaces, the displacement boundary condition is implemented at each of the center points of the holes with the following values: $U1 = \pm d/2$, $U2 = U3 = 0$.

PARTITIONING AND MESH GENERATION:

For the undertaking, hexahedral elements were taken into consideration due to their ability to produce more precise solutions with the same number of elements as tetrahedral meshes.[1] Any grid generation using a structured mesh has pros and cons. A mesh like this allows elemental cell sites to be easily addressed by double or triple indices (i,j,k) in two or three dimensions. Programming is easy with structured mesh's regular connectivity and data handling.[2] To structure the mesh the geometry was partitioned into 8 segments. For the purposes of minimizing computational spending while stimulating mesh convergence, the geometric order has been approximated quadratically, and the solution procedure is reduced integration.[3] To generate different mesh sizes the global seed size was varied ranging from 0.6 to 0.355.

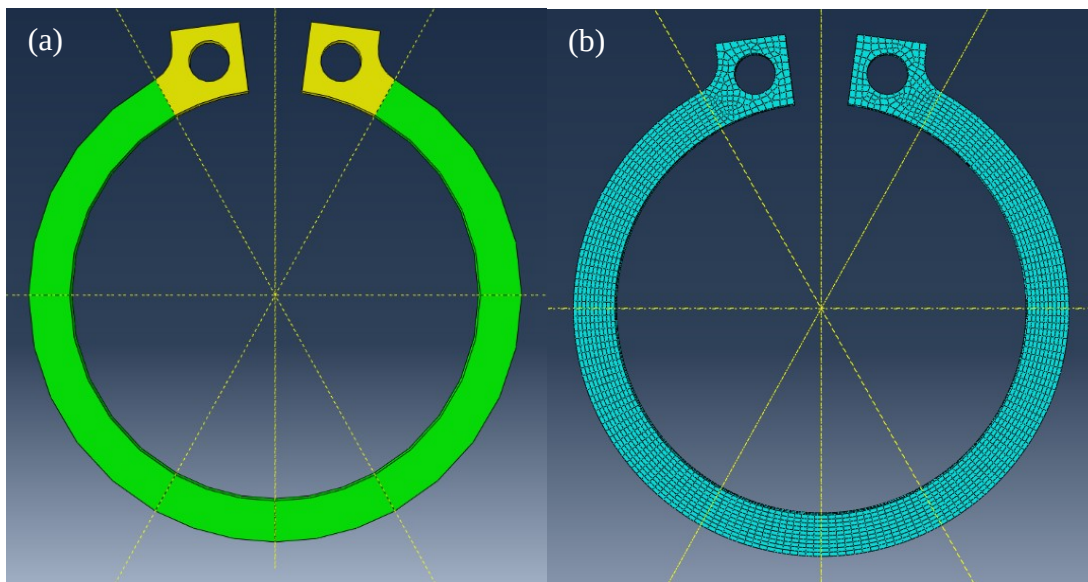


Figure 03: (a) Partitioned geometry of retaining ring created for structured mesh

(b) Structured mesh with hexahedral elements

An analysis of mesh convergence was performed for each material type, and the safety factor associated with it was assessed.

SUITABLE MATERIAL SELECTION:

Reviewing various articles, four frequently used materials for retaining rings were considered for the analysis. The maximum Von Mises Stress was taken from each iteration and dividing Yield Strength with it gave the factor of safety.

SL	Material	Mass Density (Tonnes/ mm ³)	Modulus of Elasticity (MPa)	Poisson's Ratio	Yield Strength Y (MPa)	Reference
1	Beryllium Copper, UNS C17200, TH04 Temper Strip	8.25X10 ⁻⁰⁹	130,000	0.3	1,100	[4]
2	Hard Drawn Carbon Steel ASTM-A227	7.7 X10 ⁻⁰⁹	190,000	0.27	1,034	[5]
3	SS 302	8.03 X10 ⁻⁰⁹	193,000	0.27	275	[6]
4	AISI 1060 Steel, oil quenched 845°C	7.87 X10 ⁻⁰⁹	200,000	0.29	677	[7]

MESH CONVERGANCE STUDY:

To get accurate ABAQUS simulation results, we utilize a polished mesh. Rough meshes can produce erroneous results. As mesh density increases, our model's numerical solution will become unique. Refinement of the mesh increases simulation computer resources. When mesh refinement has little effect on the solution, the mesh is converged. [8]

From the analysis it is seen that for every material the mesh converges at different number of elements.

CALCULATION AND RESULTS:

The Von Mises Stress found from the analysis was used to find factor of safety with the following formula:

For a material to be considered as suitable for use it must have a factor of safety above 1. So, from the above formula the factor of safety for each material is tabulated which is the deciding factor for the material's acceptance.

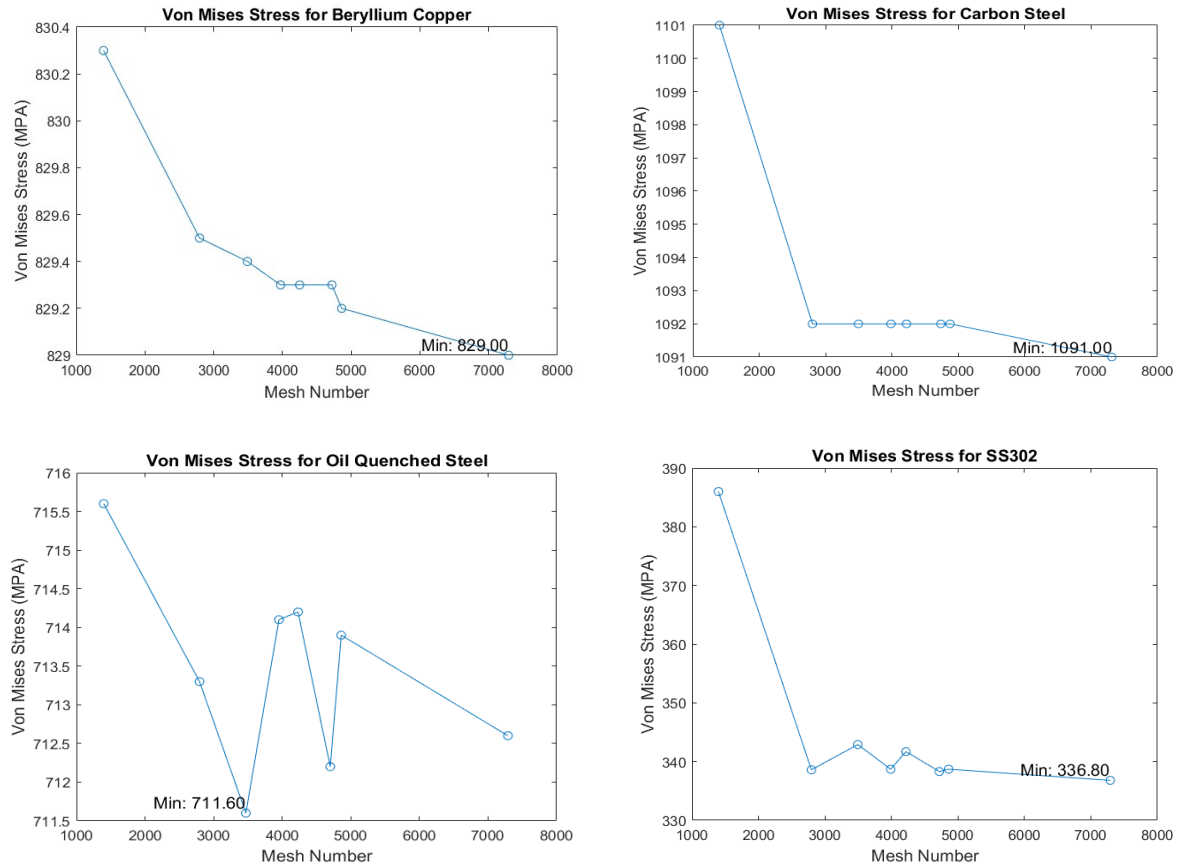


Figure 04: Von Mises Stress for varying mesh numbers marking the minimum value of stress

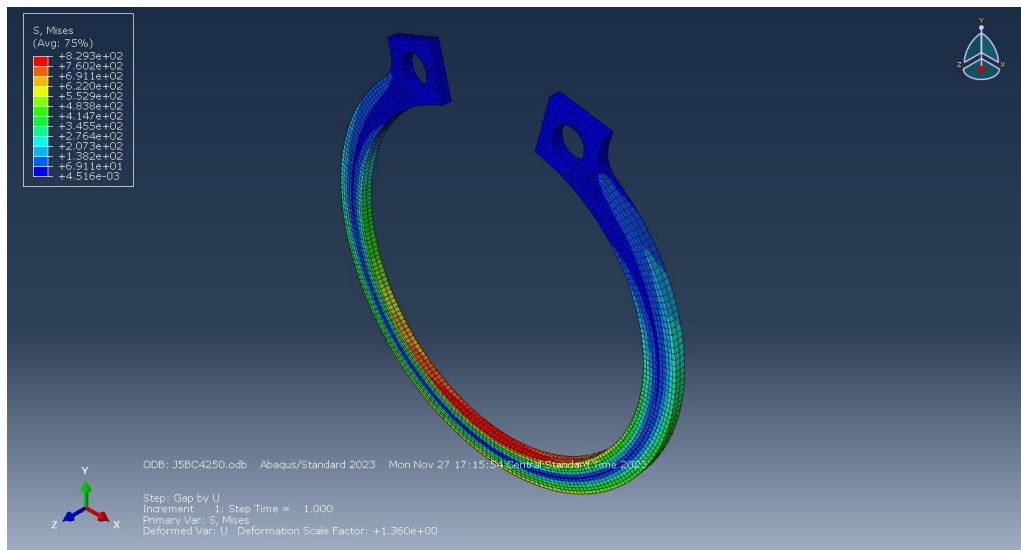


Figure 05: Maximum Stress visualization for Beryllium Copper for 4250 Elements

The analyses show that the maximum stress is generated at the bottom part of the ring. So, it is likely that yielding will occur at that point and this will loosen the fit of the material.

TABLE 01: ANALYSIS DATA FOR EACH MATERIAL WITH MESH CONVERGENCE HIGHLIGHTED			
Material	Mesh Number	V Mises Stress (MPa)	Factor of Safety Y/VM
Beryllium Copper	1397	830.30	1.325
	2794	829.50	1.326
	3490	829.40	1.326
	3972	829.30	1.326
	4250	829.30	1.326
	4720	829.30	1.326
	4860	829.20	1.327
	7290	829.00	1.327
Material	Mesh Number	V Mises Stress (MPa)	Factor of Safety Y/VM
Carbon Steel	1401	1101.00	0.939
	2802	1092.00	0.947
	3494	1092.00	0.947
	3984	1092.00	0.947
	4218	1092.00	0.947
	4734	1092.00	0.947
	4874	1092.00	0.947
	7311	1091.00	0.948
Material	Mesh Number	V Mises Stress (MPa)	Factor of Safety Y/VM
SS 302	1398	386.00	0.712
	2796	338.60	0.812
	3494	342.90	0.802
	3988	338.70	0.812
	4218	341.70	0.805
	4720	338.30	0.813
	4860	338.70	0.812
	7290	336.80	0.817
Material	Mesh Number	V Mises Stress (MPa)	Factor of Safety Y/VM
Oil Quenched Steel	1397	715.60	0.946
	2794	713.30	0.949
	3470	711.60	0.951
	3952	714.10	0.948
	4230	714.20	0.948
	4700	712.20	0.951
	4860	713.90	0.948
	7290	712.60	0.950

CONCLUSION:

From the above data Beryllium Copper (UNS C17200, TH04 Temper Strip) only meets the given criteria i.e., has significant factor of safety. So only Beryllium Copper can be considered as the suitable material for existing design of retaining ring given.

REFERENCE:

- [1] R. Biswas and R. C. Strawn, “Tetrahedral and hexahedral mesh adaptation for CFD problems,” *Applied Numerical Mathematics*, vol. 26, no. 1–2, 1998, doi: 10.1016/S0168-9274(97)00092-5.
- [2] J. Tu, G.-H. Yeoh, and C. Liu, “CFD Mesh Generation: A Practical Guideline,” *Computational Fluid Dynamics*, pp. 125–154, 2018, doi: 10.1016/B978-0-08-101127-0.00004-0.
- [3] G. Ferté, P. Massin, and N. Moës, “Convergence analysis of linear or quadratic X-FEM for curved free boundaries,” *Comput Methods Appl Mech Eng*, vol. 278, pp. 794–827, Aug. 2014, doi: 10.1016/J.CMA.2014.05.025.
- [4] https://en.wikipedia.org/wiki/Retaining_ring
- [5] <https://www.steelss.com/Carbon-steel/astm-a227.html>
- [6] <https://www.azom.com/article.aspx?ArticleID=8197>
- [7] [https://www.matweb.com/search/datasheet.aspx?
matguid=fe7b3b4ca8894c8398ab9063e89ec0b7&ckck=1](https://www.matweb.com/search/datasheet.aspx?matguid=fe7b3b4ca8894c8398ab9063e89ec0b7&ckck=1)
- [8] [https://classes.engineering.wustl.edu/2009/spring/mase5513/abaqus/docs/v6.5/books/gss/
default.htm?startat=ch04s04.html](https://classes.engineering.wustl.edu/2009/spring/mase5513/abaqus/docs/v6.5/books/gss/default.htm?startat=ch04s04.html)